

Error propagation in spatial length–weight models with implications for stock assessments

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Abstract

In fisheries management, there has been a great deal of effort focused on uncertainty quantification, especially for biomass indices used in stock assessments. Notwithstanding, many models still treat total biomass in an observed tow as a single observation ignoring the impact of preliminary analyses and/or modelling required to estimate it. Our work aims to assess the impact of propagating uncertainties in these data through to the biomass index by way of transformation of variables and nonparametric bootstrapping. We assess our proposed approaches with six different length–weight models for the sea scallop population of Georges Bank, as well as via simulation. Our findings confirm that data aggregation without proper uncertainty quantification and error propagation can substantially inflate confidence intervals for biomass indices. We make available easily implementable approaches that can improve the statistical efficiency of indices for a wide range of fisheries.

Key words: length–weight models, error propagation, spatial modelling, biomass index, bootstrap

1. Introduction

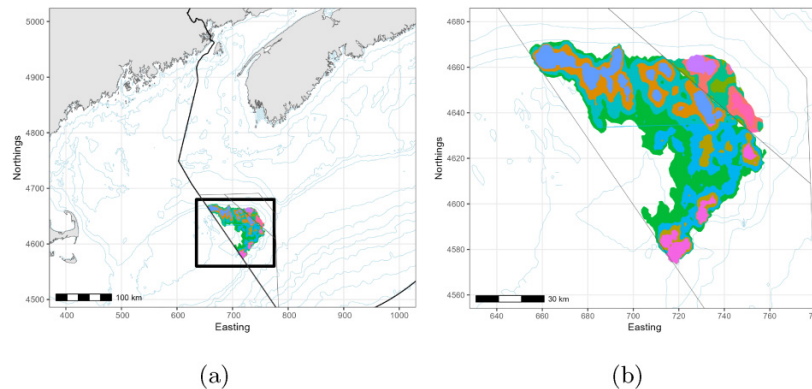
The aim of the stock assessment process is to gather all available information about a commercially exploited fish stock for the purpose of providing advice to help inform management decisions. Science advice relies heavily on statistical stock assessment models (e.g., [Beverton and Holt 1957](#); [Pella and Tomlinson 1969](#); [Feenstra et al. 2017](#); [Winker et al. 2020](#); [Nielsen et al. 2021](#)) developed to obtain estimates of population biomass/abundance and/or statistical models to estimate indices of biomass/abundance and associated measures of uncertainty ([Smith 1996](#); [Kimura and Somerton 2006](#); [Smith and Lundy 2006](#); [Gwinn et al. 2019](#); [Hansell et al. 2020](#)).

Measures of uncertainty are important components of science advice, as they can directly influence management decisions (e.g., probability of being above/below certain reference points, probability of population increasing/decreasing in the follow year). However, obtaining accurate estimates of uncertainty is often not straightforward. Fisheries data often reflect nonlinear and non-Gaussian relationships ([De Valpine 2002](#); [Jensen et al. 2005](#); [Winker et al. 2020](#)) requiring complex models with the data sources they integrate often being contradictory and difficult to capture ([Hilborn 1992](#); [Hutchings and Myers 1994](#); [Maunder et al. 2006](#); [Algave et al. 2022](#)). Aggregating these sometimes disparate datasets into yearly estimates of population status often necessitates the use of complex modelling frameworks such as index standardization models ([Thorson and Barnett 2017](#); [Zhou et al. 2019](#); [Maunder et al. 2020](#); [Luo et al. 2022](#); [Yalcin et al. 2023](#)).

One of the advantages of recent index standardization models is that they provide reliable measures of uncertainty through innate error propagation or improved model specification ([Thorson and Barnett 2017](#); [Luo et al. 2022](#); [Yalcin et al. 2023](#)), which can then be propagated into subsequent modelling steps. Error propagation methods have received a lot of attention in other fields ([Girard 2004](#); [Yen et al. 2014](#); [Bachi et al. 2021](#); [Zhang 2021](#)) and stock assessment models are increasingly built with the aim of capturing and/or propagating these uncertainties into final population estimates (e.g., estimating index uncertainty in state-space models, [Nielsen and Berg 2014](#); [Aeberhard et al. 2018](#), incorporating the index uncertainty within Bayesian priors, [Yin et al. 2019](#), etc.). While these approaches have improved stock assessments, there is still further work to be done.

Information obtained from resource assessment surveys typically represent a primary source of data that populate stock assessment models. However, it is often impossible to measure all fish in a survey tow, especially for small species. In these cases, it is common to only measure a subsample and assume it is representative of the whole population ([Nasmith et al. 2016](#); [Hardie et al. 2018](#); [DFO 2024](#)). Length–weight (LW) models, often based on a cubic relationship ([Froese 2006](#)), are fitted to these subsamples and then used to predict the weight of all other sampled fish. These weights are then summed up to obtain a total “observed” biomass for each tow. However, most index standardization models ([Thorson and Barnett 2017](#); [Luo et al. 2022](#); [Yalcin et al. 2023](#)) or design-based estimators ([Smith 1996](#); [Kimura and Somerton 2006](#))

Fig. 1. Canadian portion of Georges Bank. Black line delineates the border of the Canadian Exclusive Economic Zone, grey lines delineate management areas, blue lines are bathymetry contours, and coloured polygons are the different strata utilized for the survey. Projection system in UTM (km).



assume that these biomass estimates are actual data without associated uncertainties. Doing so may lead to inaccurate indices and associated uncertainty metrics, which can have serious consequences for fisheries management.

Our goal here is to develop a reliable biomass index for the Georges Bank sea scallop (*Placopecten magellanicus*) that takes into account all known sources of uncertainty. To achieve this goal, we test six different LW methods to determine which provides the most reliable biomass predictions. We compare three aggregation approaches for quantifying the accompanying uncertainty, and run a simulation experiment to assess their accuracy and precision. We conclude with some broader insights into appropriate methods for both uncertainty quantification and error propagation when working with LW data.

2. Methods

2.1. Data description

The Canadian section of Georges Bank is located within Scallop Fishing Area 27, an offshore scallop management area south-west of Nova Scotia, Canada (Fig. 1), and managed by Fisheries and Oceans Canada (DFO). Science advice is provided using a modified version of a delay difference biomass dynamics model (Deriso 1980; Schnute 1985; Smith and Lundy 2002; Hubley et al. 2013) that requires annual biomass indices. Both areas are surveyed annually in the summer (usually August) using stratified random sampling designs (Hubley et al. 2013). The survey gear consists of a 2.44 m wide New Bedford style offshore dredge with a 75 mm ring size and a 37 mm mesh polypropylene netting (Hubley et al. 2013).

Outside of rare extreme recruitment events wherein only a portion is used, all scallops in a tow are counted at sea and sorted into 5 mm bins. Only a subsample of live scallops (3 per 5 mm bins that are 50 mm and larger) in about half the tows have meat weights (weight of the adductor muscle) and shell heights measured (to the nearest mm), hereafter referred to as detailed sampling (Glass 2017). This subsample is then used to estimate a LW relationship every year, which is then used to predict the meat weights of all other scallops in the sam-

ple. Here, we only focus on scallops that are large enough to be the primary target of the fishery; these “fully recruited” individuals are defined as having a shell height greater than or equal to 95 mm. Data from the 2023 survey were used for this analysis, which included data from 234 tows with 82 343 captured scallops. Detailed sampling was undertaken for 115 tows with a total of 5594 scallops having their meat weight and shell height individually measured.

2.2. Model descriptions

2.2.1. LW models

All models used here are based on the following classical LW relationship (see Froese 2006 for details):

$$(1) \quad W_i = aL_i^b$$

where W_i is the weight of individual i , L_i is the length, b is called the allometric parameter, and a is an intercept (on the log scale) representing animal condition (Froese 2006). This relationship can be linearized and some stochasticity introduced such that

$$(2) \quad \log(W_i) = \log(a) + b * \log(L_i) + \varepsilon_i$$

where $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$.

We explore six different formulations of this base model. The first formulation, referred to as the *Off* method, is currently used to model scallop condition and meat weight for the offshore scallop assessment (Hubley et al. 2013). The *Off* method is a two-step modelling approach wherein the first step uses a linear mixed effect model (LMM):

$$(3) \quad W_i = a + \alpha_{\text{tow}} * L_i^b + \varepsilon_i$$

where $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$, $\alpha_{\text{tow}} \sim N(0, \sigma_\alpha^2)$ are tow-specific random effects, and b is fixed to 3 meaning that only σ_ε^2 , σ_α^2 , and a are estimated. The meat weight of all scallops in tows with detailed sampling can be predicted in this step.

Table 1. Summaries of length–weight models.

Name	Type	Error distribution	Random effect
Off	LMM + GAM	Gaussian (Both)	Independent Gaussian on L_i^b (LMM)
Off LN	LMM + GAM	Lognormal (LMM) and Gaussian (GAM)	Independent Gaussian on a (LMM)
Depth	LMM	Lognormal	N/A
Spatial	LMM	Lognormal	GRF on a
Spatial b	LMM	Lognormal	GRF on b
Spatial Both	LMM	Lognormal	GRF on both a and b

Note: LMM, linear mixed effect model; GAM, generalized additive model; GRF, Gaussian random field.

The second step uses the estimated value of a , $b = 3$ and the tow-specific α_{tow} to estimate the scallop condition (hereafter referred to as SC and related to parameter a) of a 100 mm scallop in all tows with subsampled scallops. Once these conditions are calculated, we treat them as data and model them with a generalized additive model (GAM) to obtain a relationship that can be used to predict the condition in tows without subsamples:

$$(4) \quad \text{SC}_{\text{tow}} = f(x_{\text{tow}}, y_{\text{tow}}) + f(d_{\text{tow}}) + v_{\text{tow}}$$

where SC_{tow} is the condition for a given tow, x_{tow} and y_{tow} are longitudes and latitudes, d_{tow} is the depth in metres, $f(\cdot)$ are smoothing functions using splines, and $v_{\text{tow}} \sim N(0, \sigma_v^2)$. From this relationship, we can then predict a condition in all unsampled tows using their locations and depths, and then predict the meat weight of all these unsampled scallops.

The second formulation, referred to as the *Off LN* method, also utilizes this two-model approach but with the LW relationship fitted on the log scale and b still fixed at 3. This is because the use of normally distributed errors would allow for negative values for meat weights, which is not sensible. This LW model is therefore

$$(5) \quad \log(W_i) = \log(a + \alpha_{\text{tow}}) + b * \log(L_i) + \varepsilon_i$$

The third formulation, referred to as the *Depth* method, does not include random effects but has depth-specific intercepts:

$$(6) \quad \log(W_i) = \beta * \log(d_{\text{tow}}) + b * \log(L_i) + \varepsilon_i$$

where β is the coefficient associated with depth d in a given tow. Unlike both the *Off* and *Off LN* methods, b is estimated and no GAM is required to predict the meat weights in tows without detailed sampling as the depths for those tows are also known.

The fourth formulation, referred to as the *Spatial* method, is highly similar to the third but adds a spatial random effect α_s at location s :

$$(7) \quad \log(W_i) = \beta * \log(d_{\text{tow}}) + \alpha_s + b * \log(L_i) + \varepsilon_i$$

These spatial random effects are modelled as a Gaussian random field (GRF) with mean zero and covariance matrix $\Sigma(s, s') = \text{Matérn}(s - s')$, indicating that the covariance between two locations s and s' where $s \neq s'$ follows an isotropic

Matérn covariance:

$$(8) \quad \text{Matérn}(s - s') : \sigma_\alpha^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{d(s, s')}{\rho} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{d(s, s')}{\rho} \right)$$

where Γ is the gamma function, K_ν is the modified Bessel function of the second kind, ρ is the range parameter, σ_α^2 is the spatial variance, and ν is the smoothness parameter fixed at 1.

The fifth formulation, referred to as the *Spatial b* method, takes a very similar formulation to eq. 7 but the GRF only applies to the allometric parameter b :

$$(9) \quad \log(W_i) = \beta * \log(d_{\text{tow}}) + (\alpha_s + b) * \log(L_i) + \varepsilon_i$$

which constrains the spatial variability to only be related to b .

The last formulation, referred to as the *Spatial Both* method, puts a separate random field on both the intercept and slope:

$$(10) \quad \log(W_i) = \beta * \log(d_{\text{tow}}) + \omega_s + (\alpha_s + b) * \log(L_i) + \varepsilon_i$$

where ω_s is the value of the intercept-specific GRF, allowing both the intercept and the slope parameter b to vary separately. All six formulations are summarized in Table 1.

2.2.2. Biomass index and confidence intervals

We now scale the “observed” biomass in each tow to the surveyed area of Georges Bank and take the mean of these scaled biomasses to obtain a biomass index. Three different approaches are used for uncertainty (95% CIs). The first, which we refer to as the *Straight Mean* approach, makes the traditional assumption that the tow-specific total biomasses are directly observed and is simply calculated using the formula for the variance of the mean, although the estimates are also based on the models described in the previous section and therefore utilize individual-level data like the other approaches. This approach does not attempt to propagate uncertainties. For the second approach (*Direct CI*), uncertainty is propagated through transformation of variables. Using eq. 1 as an example and since we assume L_i , then calculating the variance of $\log(W_i)$ (and therefore of W_i) is trivial since $\text{Cov}(\log(a), \log(L_i)) = \log(L_i) \text{Cov}(\log(a), b)$, assuming asymptotic normality for parameters a and b and using the uncertainties estimated while modeling as their variance. From

this we can then use the classical results of $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$ and similar results for more than two random variables (see Supplementary Materials for more details). For the final approach (*Bootstrapped CI*), uncertainty is propagated through a nonparametric bootstrap (percentile method, Efron 1982) due to its easy implementation and attractive statistical properties (Efron 1982; Meyer et al. 1986). A relatively small bootstrap sample size of 1000 might produce imperfect coverage of the distribution tails, but was chosen due to the very large computational time required for bootstrapping multiple spatial models. Results were fairly stable.

A challenge in obtaining these indices resides with the fact that a sum of lognormal random variables (i.e., predicted meat weights) does not have a standard distribution (Lam and Le-Ngoc 2006). To approximate this distribution, we utilize the method developed by Lo (2013) and available in the *lognorm* package in R. It assumes that the sum of multiple lognormal variables X_i can be approximated in the following way:

$$(11) \quad \mu_S = \ln(S_+) - \sigma_S^2/2$$

where $S_+ = \sum_i E[X_i]$ is the expected value of the sum of lognormal variables, μ_S and σ_S^2 are the lognormal distribution parameters of the sum of lognormal variables where $\sigma_S^2 = \frac{1}{S_+^2} \sum_{ij} \text{cor}_{ij} \sigma_i \sigma_j E[X_i] E[X_j]$, μ_i and σ_i^2 are the lognormal distribution parameters of the individual variables, and cor_{ij} is the correlation between two random variables i and j on the log scale.

2.2.3. Case study

All six LW methods are fitted to the 2023 Georges Bank sea scallop data from the DFO survey. Model performance is assessed based on standard LM and LMM methods (i.e., residual structure, quantile–quantile plots, etc.) alongside two types of cross-validation (CV): 10-fold CV, and leave-one-group-out (LOGO) CV. The 10-fold CV is used to identify which method performs best when predicting within each tow with detailed sampling as it simply predicts on the missing folds. However, it does not tell us anything about which method is best at out-of-sample predictions. To determine which method best predicts the LW relationship for unsampled tows, LOGO CV is utilized. This procedure iteratively removes all data from individual tows and predicts the results for the excluded data. Lastly, we use Akaike information criterion (AIC), Bayesian information criterion (BIC), and conditional AIC (Zheng et al. 2024), which is more appropriate for models with random effects, to further support model selection.

We use the midpoint of each size bin as the shell height used for predictions (alternatives presented in the Supplementary Materials). All models are fit within a maximum likelihood framework using the Template Model Builder (TMB) package in R (Kristensen et al. 2016) due to its computational efficiency and high accuracy, with the exception of the GAMs

Table 2. Root mean square prediction error for all length–weight models.

Model	10-Fold CV	LOGO CV
Off	3.272	5.421
Off LN	3.292	5.386
Depth LN	4.541	6.791
Spatial	3.317	5.739
Spatial <i>b</i>	4.050	6.275
Spatial Both	3.258	5.737

Note: Note that for the Off and Off LN method, the 10-fold cross-validation (CV) is based on the linear mixed effect model part of the method while the leave-one-group-out (LOGO) CV is based on the generalized additive model part.

in both *Off* approaches, which are fit using the *mgcv* package (Wood 2006).

2.2.4. Simulation experiment

A simulation experiment is carried out to assess which of the two CI approaches that do error propagation is most accurate and representative of its nominal coverage. The data for the simulations are generated using the *Spatial* method as it was identified as one of the most appropriate in preliminary analyses. We simulate data based on the conditions seen for the case study, that is to say 234 tows with a total of 82 343 scallops for the index, and 115 tows with a total of 5594 scallops for the detailed sampling for the LW models. The parameters chosen for the simulation experiment are the same as those estimated for the real data, and instead of simulating a completely new random field we utilize the one predicted in our case study so as to mimic our real life setting and as is often done in this type of simulation (Zheng and Cadigan 2023). All methods are then used with the simulated data to calculate an index with two CIs (*Direct CI* and *Bootstrapped CI*). The bootstrap simulation is limited to 200 datasets (each of which is bootstrapped 1000 times) due to the computational demands of fitting multiple spatial models.

3. Results

3.1. Case study

3.1.1. LW models

All methods successfully converged, and model assumptions appear to be respected except for the *Off* method due to the inappropriate use of a normal distribution for its error terms. The nonspatial models show some minor spatial patterns in their residuals (see Supplementary Material for details).

The prediction accuracy of the methods with either spatial or independent Gaussian random effects have similar root mean square prediction error (RMSPE) and outperform the *Depth* method, although the *Spatial b* method performance is worse than the other random effects models (Table 2). This is the case for both the 10-fold and LOGO CV procedures, and the best methods are the *Off*, *Off LN*, *Spatial*, and *Spatial*

Fig. 2. Root mean square prediction error (RMSPE) from 10-fold cross-validation in every tow with detailed sampling. Projection system in UTM (km).

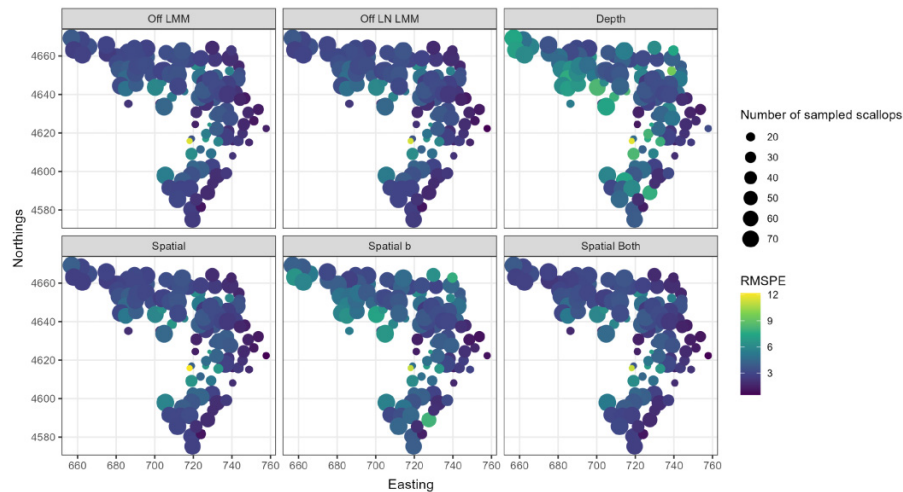


Table 3. Akaike information criterion (AIC), Bayesian information criterion (BIC), and cAIC (conditional AIC) for all one-step length-weight method (since the two-step methods would have two of each information criterion and making comparisons difficult).

Model	AIC	BIC	cAIC
Depth LN	-2206	-2184	N/A
Spatial	-6257	-6224	-5279
Spatial <i>b</i>	-3186	-3153	-2635
Spatial Both	-7177	-7131	-5524

Note: Note that cAIC is only appropriate for models with random effects.

Both methods that all appear roughly equivalent. Spatially, all methods have similar patterns of RMSPE, with the inside portion of the bank generally seeing higher RMSPE associated with lower sample sizes, while the *Spatial b* method has higher RMSPE on the eastern edge (Fig. 2). The LOGO CV results in higher RMSPE everywhere with the spatial pattern being similar to the 10-fold CV figure (see Supplementary Materials for LOGO CV results). All three information criterion (AIC, BIC, and cAIC) favour *Spatial Both* as it has the lowest value for all three (Table 3).

3.1.2. Index confidence intervals approaches

Using the *Straight Mean* approach and not propagating uncertainties results in substantially larger CIs than when uncertainty-propagation methods are used (Fig. 3). However, there are noticeable differences between the *Bootstrapped CIs* and *Direct CIs* approaches for all methods except the *Spatial* method, which is the only one in which both CIs are almost identical. The two *Off* methods have substantially smaller bootstrapped CIs compared to the *Direct CIs*, these last ones being much larger than for the other LW methods. The *Depth* method has a smaller *Direct CI* than *Bootstrapped CI*. Finally,

while both types of CIs are similar in size for the *Spatial Both* and *Spatial b* methods, there is a slight but noticeable increase in the size of the CI using the *Direct CI* approach compared to the *Bootstrapped CI* approach.

3.2. Simulation experiment

The case study indicates that the three best performing methods in terms of diagnostics and RMSPE were the *Off LN*, *Spatial*, and *Spatial Both* methods. For this reason, we focus the simulation results on those, with the results for the other models available in the Supplementary Materials.

The only combination of methods and CI approaches to achieve the expected coverage rate is the *Bootstrapped CI* for the *Spatial Both* method (Table 4). However, the CI for the *Direct CI* approach for the same method is overly conservative. The coverage of the *Spatial* method, which was used to simulate the data, is slightly lower than expected for both CI approaches (91.5%).

While the *Spatial Both* method has the coverage closest to the expected 95%, the distribution of estimates using this method are consistently positively biased while the *Off LN* method is consistently negatively biased, with neither methods ever overlapping the correct value (Fig. 4). The correct method, the *Spatial* method (as it is the one used to simulate the data), is the only one without any indication of bias, although it has a wider distribution of estimates than the *Spatial Both* method.

4. Discussion

We found a substantial reduction in the size of CIs by incorporating the raw data directly, and the errors therein, rather than using aggregated tow-specific sums assumed to have been directly observed, i.e., without errors. The increased sample size (instead of 234 tows, the sample size is 5594 scallops) and the efficiency gains from modelling individual data points rather than sums and means thereof are responsible

Fig. 3. Estimated indices with CIs calculated through bootstrapping, direct calculation, and based on treating the tow-specific sums as direct observations for all models. LW, length-weight.

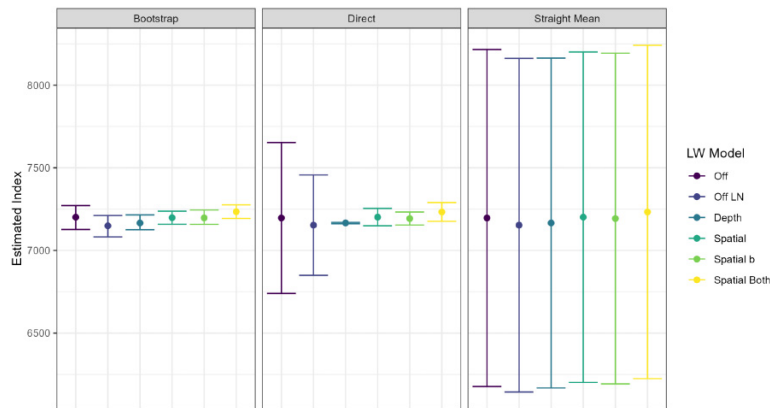


Table 4. Nominal percent coverage of bootstrapped and direct CIs for abundance index obtained from the *Off LN*, the *Spatial*, and the *Spatial Both* methods.

Model	Bootstrapped CI (%)	Direct CI (%)
Off LN	76.5	N/A
Spatial	91.5	91.5
Spatial Both	95.5	100

Note: Note that for the *Off LN* method, only the bootstrapped CIs are calculated.

for this reduction in uncertainty. This suggests a relatively straightforward gain in efficiency easily available for most fisheries assessments that could lead to more reliable survey indices.

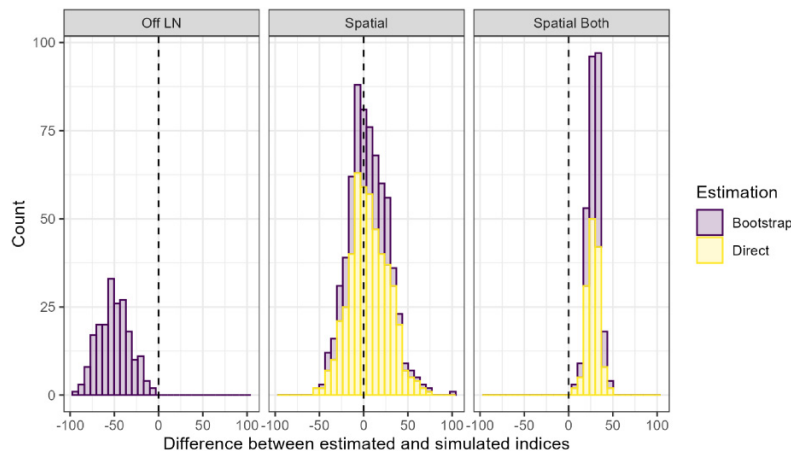
For selecting the most appropriate model, fisheries science has provided varying advice, whether favouring parsimony (Thorson 2019) or flexibility (Bolker et al. 2009). There is therefore an argument for selecting both the *Spatial* or *Spatial Both* method. Given their similar diagnostics and predictive capabilities, the main difference is their performance with the two different CI approaches. For a correctly specified model with a large sample size, theoretical *Direct CIs* and nonparametric *Bootstrapped CIs* should be similar in size as confirmed by our simulations. A loss of efficiency in the *Direct CIs* in our case study for the *Spatial Both* model might therefore be an indication of model misspecification, pointing toward the *Spatial* method as the more accurate method. This is further supported by the simulations results mimicking the case study results when setting the *Spatial* method as the data-generating model. However, this is contradicted by the superior performance of the *Spatial Both* method in terms of AIC, BIC, and cAIC, meaning that guidance around model selection might instead suggest the *Spatial Both* as the appropriate method.

Irrespective of the model selected, incorporating spatial random effects substantially improved results according to every measure considered in this study (diagnostics, RMSPE, AIC/BIC). These results substantiate earlier recommendations to account for spatial dynamics within stock assessment pro-

cesses (Punt 2019). Much of the effort to date has focused on incorporating spatial dynamics directly into the stock assessment models (e.g., Cadigan et al. 2017; McDonald et al. 2021), but many other components of stock assessments (growth rates, age-length curves, etc.) can also benefit from the inclusion of spatial dynamics (e.g., Cahill et al. 2020; Yin et al. 2022). Even if the stock assessment model is itself nonspatial, accounting for spatial dynamics in the development of the indices feeding the assessment model can result in improved model estimates and reduced uncertainty (Thorson et al. 2015). The spatial LW models used here can help identify spatial differences in LW relations, which, as our simulations show, can help reduce bias in both the LW relationship and in biomass-based growth rates that rely on LW models for their estimation (e.g., Nasmith et al. 2016). Furthermore, recent computational advances such as the development of sdmTMB (Anderson et al. 2022) and the starve package (Lawler et al. 2023) have greatly reduced the complexity of developing spatial models and make it relatively straightforward to operationalize these methods within existing stock assessment frameworks.

Moreover, these methods would benefit any stock assessment that uses design-based estimation to obtain survey indices in their preliminary steps. Indices developed from stratified surveys rely on the underlying strata properly representing the variability in the population being monitored and aggregating areas that tend to be mostly homogeneous regarding the variable of interest (Särndal 1978). However, this is an often challenging process, and many fisheries utilize simpler method to define their strata (e.g., Trzcinski et al. 2009; Hardie et al. 2018). Using spatial models to standardize catch rates (e.g., Luo et al. 2022) should account for spatial variability in the ecosystem and incorporate observed variability in stock productivity. These spatial variables (e.g., spatial growth rates) can then be used to delineate homogeneous areas and improve the identification of these strata. Building on this approach, one could then extend it to identify these homogeneous areas to support the delineation of larger stock units based on biologically-meaningful information (Cadrin 2020).

Fig. 4. Histograms of differences between estimated and simulated indices from the output of the *Spatial* and *Spatial Both* models. Dashed lines represent no difference.



These findings reflect a similar conclusion to other analyses incorporating observation uncertainties within stock assessments (McDonald et al. 2022; Breivik et al. 2024): ignoring preliminary modelling and analyses can introduce unnecessary uncertainty into the stock assessment process. The analyst should always keep in mind the nature of the data generating process and the nature of the actual data selected in the observed sample so as to properly analyze it. In our example, the sample is not the tow-specific “observed” biomasses, but rather measurements of all individual captured animals from which all subsequent analyses derive.

Our findings suggest confidence intervals for indices in multiple fisheries may actually be smaller than previously thought and their uncertainty could benefit from a reassessment. Specifically for the Georges Bank sea scallop assessment, which currently utilizes a Bayesian model that estimates index variances as captured by credible intervals (Jonsen et al. 2009; Hubley et al. 2014), our approach results in a more reliable index forming the basis for the science advice provided to management. More generally, this type of analysis could be undertaken for multiple other components of stock assessments and potentially further improve the uncertainty quantification for analyses including the estimation of age-length curves (e.g., Trzcinski et al. 2009), of yearly growth parameters (e.g., Koeller 2006; Maunder et al. 2016), or the inclusion of externally-predicted covariates increasingly used to include the impact of the environment (Pedersen et al. 2022).

However, one should keep in mind that there are various ways in which an index could be unrepresentative of the actual population dynamics. These include but are not limited to space- and time-varying catchability (Thorson 2011; Kotwicki and Ono 2019), confounding between various processes (e.g., between selectivity, growth and fishing mortality (Maunder et al. 2016), and population movements (Tanaka et al. 2020)). In addition to these challenges, even when sample sizes appear relatively large, the proportion of the area that these surveys cover is often extremely small. Most standard statistical methods cannot properly account for sam-

pling that might be unrepresentative and assessing how representative a given sample is in an environment as uncertain as the ocean is exceedingly difficult.

While reduced uncertainty may suggest the index well reflects what was observed within a given survey, it also puts into perspective the large number of unknown complex ecological factors impacting both the sampling process and the underlying population dynamics. There are no easy solutions here, although the recent growing interest in new survey methods such as towed or drop cameras (Stokesbury and Bethoney 2020) might, in conjunction with traditional surveys (essential for stock assessment components such as growth, aging, etc.), improve the ability of surveys to cover more space and more closely track the overall population health. Thus, while these methods reduce the uncertainty by providing statistically sound estimates, the practitioner should remain cognizant of other factors (e.g., poor survey design) that could impact their confidence in the indicator but cannot currently be captured using these techniques.

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Data availability

The anonymized data, simulation scripts, and model scripts are available at <https://github.com/RaphMcDo/Error-Prop-Scripts> (<https://doi.org/10.5281/zenodo.14896315>).

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Methodology: RRM, DMK, JMF

Project administration: RRM, DMK, JMF

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Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfas-2024-0345>.

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